

Tarea No. 3

3.55

Distancia:

Componentes

$$V_{ox} = 90 \text{ m/s} \cos(23^\circ) = 82.85 \text{ m/s}$$

$$V_{oy} = 90 \text{ m/s} \sin(23^\circ) = 35.77 \text{ m/s}$$

$$y_f = 0, \quad y_0 = h = 174 \text{ m}$$

Tiempo:

$$y_f = y_0 + V_{oy}t - \frac{gt^2}{2} = 0 \text{ m} = 174 + 35.77t - \frac{9.8t^2}{2}$$

Cuadrática =

$$x = V_{ox}t = 82.85(9.59)$$

$$x = 795 \text{ m}$$

3.59 El jinroón más largo

$$a) \quad a_y = -9.8 \text{ m/s}^2, \quad y - y_0 = -0.9 \text{ m} \quad V_{oy} = V_0 \sin 45^\circ$$
$$y - y_0 = V_{oy}t + \frac{1}{2}a_y t^2$$

$$-0.9 = (V_0 \sin 45^\circ)t + \frac{1}{2}(-9.8)t^2$$

$$x - x_0 = V_{ox}t + \frac{1}{2}a_x t^2$$

$$t = \frac{x - x_0}{V_x} = \frac{188 \text{ m}}{V_0 \cos 45^\circ}$$

$$-0.9 = (v_0 \sin 45) \left(\frac{188}{v_0 \cos 45} \right) - (4.9) \left(\frac{188}{v_0 \cos 45} \right)^2$$

$$4.9 \left(\frac{188}{v_0 \cos 45} \right)^2 = 188 + 0.9 = 188.9 \text{ m}$$

$$\left(\frac{v_0 \cos 45}{188} \right) = \left(\frac{4.9}{188.9} \right) = v_0 = \left(\frac{188}{\cos 45} \right) \sqrt{\frac{4.9}{188.9}} = 42.8 \text{ m/s}$$

$$b) v_{0y} = v_0 \sin 45^\circ = 30.3, \quad t = 3.83 \text{ s}$$

$$y - y_0 = v_{0y} t + \frac{1}{2} a_y t^2$$

$$y - y_0 = 176 - 77.9 = 98.1$$

$$y - y_0 = (30.3)(3.83) + \frac{1}{2} (-9.8)(3.83)^2$$

$$45 - 3 = 42 \text{ m}$$

3.61

$$h = \frac{v_0^2 \sin^2 \alpha_0}{2g}$$

$$R = \frac{v_0^2 \sin 2\alpha_0}{g}$$

$$a) h = \frac{v_0^2}{2g}$$

$$y \quad v_0 = \sqrt{2gh}$$

$$b) v_0 = 2\sqrt{2gh}$$

$$h = \frac{8gh \sin^2 \alpha_0}{2g} = 4h \sin^2 \alpha_0$$

$$\sin \alpha_0 = \frac{1}{2} \quad y \quad \alpha_0 = 30^\circ$$

$$c) R = \frac{(2\sqrt{2gh})^2 \sin 60}{g} = 6.93h$$

$$3.63 a) y - y_0 = 0.0674 \text{ m}$$

$$v_y^2 = v_{0y}^2 + 2a_y(y - y_0)$$

$$v_{0y} = \sqrt{-2a_y(y - y_0)} = \sqrt{2(-9.8)(0.0674)} = 1.15 \text{ m/s}$$

$$v_{0y} = v_0 \sin \alpha_0$$

$$v_0 = \frac{v_{0y}}{\sin \alpha_0} = \frac{1.15}{\sin 50} = 1.5 \text{ m/s}$$

$$b) X - X_0 = 7.06 \text{ m} \quad X - X_0 = V_{0x}t + \frac{1}{2}a_x t^2 \quad t = \frac{X - X_0}{V_{0x}} = \frac{X - X_0}{V_0 \cos 50} = 7.105$$

$$Y - Y_0 = V_{0y}t + \frac{1}{2}a_y t^2 = (7.15)(7.10) + \frac{1}{2}(-9.8)(7.10)^2 = -4.66 \text{ m}$$

3.65 ¡Cuidado!

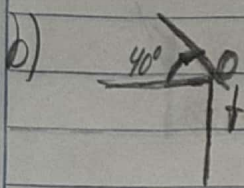
$$a_x = 0 \quad a_y = 9.8 \text{ m/s}^2 \quad V_{0x} = V_0 \cos \theta = 5.36$$

$$V_{0y} = V_0 \sin \theta = 4.5 \text{ m/s}$$

$$d) Y - Y_0 = V_{0y}t + \frac{1}{2}a_y t^2 \quad Y - Y_0 = 14 \quad 14 = 4.5t + 4.9t^2$$

$$t = 1.295$$

$$X - X_0 = V_{0x}t + \frac{1}{2}a_x t^2 = (5.36)(1.29) = 6.91$$



$$c) X - X_0 = V_{0x}t + \frac{1}{2}a_x t^2 \quad t = \frac{X - X_0}{V_{0x}} = \frac{4}{5.36} = 0.7465$$

$$Y - Y_0 = V_{0y}t + \frac{1}{2}a_y t^2 = 6.08 \text{ m}$$

$$14 - 6.08 = 7.9 \text{ m} \text{ en la altura}$$

3.67 Salto del río II

$$a) V_{0y} = V_0 \sin 53$$

$$Y - Y_0 = V_{0y}t + \frac{1}{2}a_y t^2$$

$$-75 = (V_0 \sin 53)t - 4.9t^2$$

$$V_{0x} = V_0 \cos 53$$

$$X - X_0 = V_{0x}t + \frac{1}{2}a_x t^2$$

$$40 = (V_0 \cos 53)t$$

$$V_0 = \frac{40}{\cos 53} = 66.47$$

$$-75 = 66.47 \sin 53 - 4.9t^2$$

$$t = \sqrt{\frac{66.47 \sin 53 + 75}{4.9}} = \sqrt{\frac{68.08}{4.9}} = 3.727$$

$$V_0 = \frac{40}{t \cos 53} = \frac{40}{3.727 \cos 53} = 77.8 \text{ m/s}$$

$$b) V_{0y} = V_0 \sin 53 = 7.77 \text{ m/s}$$

$$y - y_0 = V_{0y}t + \frac{1}{2}a_y t^2 \quad -100 = 7.7t - 4.9t^2$$

$$\text{cuadrática} = t = 5.30 \text{ s}$$

$$x - x_0 = V_{0x}t = V_0 \cos 53 t = (5.36)(5.3) = 28.4 \text{ m}$$

$$3.69 \ a) V_y^2 = V_{0y}^2 + 2a_y(y - y_0) \quad y - y_0 = 87.6 \text{ m}$$

b) Tienda misma velocidad horizontal, mientras el cohete está en el aire y el cohete aterriza en el carro.

$$c) V_y = V_{0y} + a_y t \quad t = 8.764 \text{ s}$$

$$x - x_0 = V_{0x}t = (30)(8.764) = 245 \text{ m}$$

$$d) V_{0x} = 30 \text{ m/s} \quad V_{0y} = 40$$

por lo tanto 53.7° por encima de la horizontal

$$3.77 \ a) y - y_0 = V_{0y}t + \frac{1}{2}a_y t^2 \quad y - y_0 = 20$$

$$t = \sqrt{\frac{2(y - y_0)}{a_y}} = \sqrt{\frac{2(20)}{9.8}} = 2.02 \text{ s}$$

$$x - x_0 = V_{0x}t + \frac{1}{2}a_x t^2 \quad V_0 = V_{0x} = \frac{x - x_0}{t} = \frac{100}{2.02} = 49.5 \text{ m/s}$$

$$b) y - y_0 = 45 \quad t = \sqrt{\frac{2(y - y_0)}{a_y}} = \sqrt{\frac{2(45)}{9.8}} = 3.03$$

$$x - x_0 = v_{0x}t = (49.5)(3.03) = 150m \quad 750 - 700 = 50m$$

$$3.73 \quad v_{0y} = v_0 \sin \alpha = 43.4 \quad a_y = -9.8 \quad y - y_0 = 0$$

$$y - y_0 = v_{0y}t + \frac{1}{2}a_y t^2 \quad t = 8.86s$$

$$\#1 \quad 246.2 + 15 = 261.2$$

$$(226.2)(8.86) = 2000m$$

$$\#2 \quad 261.2 - 35 = 226.2 \text{ durante } 8.86s \text{ que está en el aire.}$$

$$d) \text{ Separador por } = 2000m \text{ en } 8.86s$$

$$\#1 \text{ recap } 133m \text{ y el } \#2 \text{ viaja } 370m$$

$$370 - 133 = 237m \rightarrow 2.180m$$

$$3.75 \quad a_x = 0 \quad a_y = -9.8 m/s^2$$

$$v_{F/G-x} = 25 \cos 30 = 21.65 m/s$$

$$v_{A/F-x} = 20 \cos 53 = 12 m/s \quad v_{A/F-y} = 20 \sin 53 = 16 m/s$$

$$v_{A/G-x} = v_{A/F-x} + v_{F/G-x} = 12 + 21.65 = 33.7 m/s$$

$$v_{A/G-y} = v_{A/F-y} + v_{F/G-y} = 16 m/s$$

$$\tan \alpha_0 = \frac{v_{A/G-y}}{v_{A/G-x}} = \frac{16}{33.7} = \alpha_0 = 25.4^\circ \quad v_{A/F-y} = -16 m/s$$

25.4° debajo de la horizontal.

$$3.77 \quad a_x = 0 \quad a_y = -g$$

$$V_{0x} = V_0 \cos 45^\circ \quad V_{0y} = V_0 \sin 45^\circ \quad 90 \text{ km/h} = 25 \text{ m/s}$$

$$V = 30.6 - 25 = 5.6 \text{ m/s}$$

$$y - y_0 = V_{0y}t + \frac{1}{2}a_y t^2 \quad y - y_0 = 0$$

$$t = \frac{2V_{0y}}{a_y} = \frac{2V_0 \sin 45^\circ}{g} = \quad d = V_0 t = \frac{\sqrt{2} V_0^2}{g}$$

$$R = V_{0x}t = V_0 \cos 45^\circ t = \frac{2V_0^2 \sin 45^\circ \cos 45^\circ}{g} = \frac{V_0^2}{g}$$

$$R = d + 75.8 \text{ m}$$

$$\frac{V_0^2}{g} = \frac{\sqrt{2} V_0^2}{g} + 75.8$$

$$V_0^2 - 7.92 V_0 - 754.8 = 0$$

$$V_0 = 67.2 \text{ km/h}$$

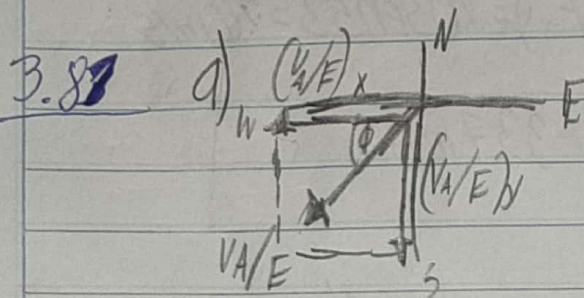
$$V_{0x} = V_0 \cos 45^\circ = 43.3 \text{ km/h}$$

$$V_{Ex} = 43.3 + 90 = 133.3 \text{ km/h}$$

$$V_{0y} = V_0 \sin 45^\circ = 43.3 \text{ km/h}$$

$$V_{Ey} = 43.3$$

$$V_{Ex} = \sqrt{V_{Ex}^2 + V_{Ey}^2} = 140 \text{ km/h}$$



$$V_{A/E} = \sqrt{(V_{A/E})_x^2 + (V_{A/E})_y^2}$$

$$V_{A/E} = 44.7 \text{ km/h}$$

$$\tan \theta = 40/20 = 2$$

$$\theta = 63.4^\circ$$

$$d) \sin \theta = \frac{V_{A/E}}{V_{P/A}} = \frac{40}{220} = 0.1818 \quad \theta = 10.5^\circ \text{ north}$$

3.83

$$V_{j/g} = 8 \text{ m/s} \quad V_{B/g} = 12 \text{ m/s}$$

$$V_{B/jx} = V_{B/g} \sin 37 = 7.222 \text{ m/s}$$

$$V_{B/jy} = V_{B/g} \cos 37 - V_{j/g} = 7.584 \text{ m/s}$$

$$V_{B/j} = 7.39 \text{ m/s} \quad 12.40 \text{ m/s}$$

3.85

$$a) \quad V_{oy} = 0 \quad y - y_0 = -3 \text{ m} \quad a_y = -13.8 \text{ m/s}^2$$

$$y - y_0 = V_{oy}t + \frac{1}{2}a_y t^2 \quad t = \sqrt{\frac{2(y - y_0)}{a_y}} = \sqrt{\frac{2(-3)}{-13.8}} = 0.659 \text{ s}$$

$$b) \quad V_y = V_{oy} + a_y t \quad V_{oy} = 0 \quad t = 0.659$$

$$i) \quad a_y = -13.8 \text{ m/s}^2 \quad V_y = -9.09 \quad 9.09 \text{ m/s}$$

$$ii) \quad a_y = -9.8 \text{ m/s}^2 \quad V_y = -6.46 \text{ m/s} \quad 6.46 \text{ m/s}$$

$$c) \quad y - y_0 = V_{oy}t + \frac{1}{2}a_y t^2$$

$$i) \quad a_y = -13.8 \text{ m/s}^2 \quad y - y_0 = \frac{1}{2}(-9.8)(0.659)^2 = 3 \text{ m}$$

$$ii) \quad y - y_0 = \frac{1}{2}(-9.8)(0.659)^2 = 2.13 \text{ m}$$

3.87

$$\tan \theta = \tan(\theta + \phi) - \left(\frac{g_x}{2v^2} \right) \frac{1}{\cos(\theta + \phi)}$$

$$g_x / 2v^2 = B$$

$$B = \frac{(9.8)(60) \cos 30}{2(32)^2} = 0.2484$$

$$\sin \theta \cos(\theta + \phi) = \sin(\theta + \phi) \cos \theta - \frac{B \cos \theta}{\cos(\theta + \phi)}$$

$$\sin(\theta + \phi) \cos \theta - \cos(\theta + \phi) \sin \theta = \frac{B \cos \theta}{\cos(\theta + \phi)}$$

$$\sin((\theta + \phi) - \theta) = \sin \theta$$

$$\sin \phi \cos(\theta + \phi) = B \cos \theta$$

$$\frac{1}{2} \sin(2\phi + \theta) + \sin \theta = B \cos \theta$$

a) $\theta = 30^\circ$ $\phi = 19.3$ $\theta + \phi = 49.3^\circ$

$B = 0.2877$ $\phi = 77.5$

b) $\theta = -30$ $\phi = 30$ $\phi = 73$

$$\cos 30 = \cos -30$$

$$\phi + \theta = -77$$